

PRD_CRM_Module

Clients, Contacts, Leads, Opportunities and Sales

Full Product Requirements Document for Product, UX, Frontend, Backend, QA, Security and DevOps Teams

The CRM Module is the relationship and revenue engine of the platform. It manages individual clients, client companies, contacts, leads, opportunities, pipelines, reservations, purchases, contracts, payments, communications, support and complete relationship history.

Global rule: every CRM page, section, widget, feature and action must support Platform Configuration. The first implementation exposes the full capability set; detailed user-type restrictions are defined later.

Status: Confirmed product scope

Version: 1.0

Audience: Full-stack implementation team

1. Document Control

Field	Value
Document	PRD_CRM_Module
Primary pages	CRM Module Page, Client Page, Contact Page, Lead Page, Opportunity Page, Pipeline Page
Connected modules	Companies, Projects, Buildings, Units, Documents, Tasks, Meetings, Approvals, Contracts, Finance, Calendar, Notifications, Users, Teams, Reports
Initial access model	Full capability first; restrictions later
Platform Configuration	Mandatory compatibility at module, page, section, widget, feature and action level
Data ownership	CRM owns client, contact, lead, opportunity and relationship records; linked modules remain source of truth for their own records

2. Product Vision

- Create one universal CRM for real estate, construction, services, retail, logistics, hospitality and other business categories.
- Support both individual clients and client companies with multiple contacts.
- Connect every commercial relationship to projects, units, contracts, payments, documents, tasks and meetings.
- Provide a configurable sales pipeline and complete client timeline.
- Avoid duplicate client and contact records.
- Support both pre-sale and after-sale operations.

CRM is not only a contact list. It is the complete relationship, sales and after-sales workspace.

3. Scope and Boundaries

- In scope: clients, contacts, companies, leads, opportunities, pipelines, reservations, purchases, contracts, payments, communications, support, referrals, marketing, reports and settings.
- In scope: responsive desktop, tablet and mobile behavior.
- In scope: data model, APIs, validation, security, audit, performance and testing.
- Out of scope: live email/WhatsApp/SMS integrations in the first phase.
- Out of scope: final permissions matrix and client portal implementation.

4. CRM Information Architecture

- CRM Dashboard
- Clients
- Client Page
- Contacts
- Contact Page
- Client Companies
- Leads
- Lead Page
- Opportunities
- Opportunity Page
- Sales Pipeline
- Reservations

- Purchases
- Units
- Contracts
- Payments
- Documents
- Tasks
- Meetings
- Communications
- Notes
- Approvals
- Timeline
- Support
- Referrals
- Marketing
- Reports
- CRM Settings

5. CRM Dashboard

- Total clients
- Active clients
- New leads
- Qualified leads
- Open opportunities
- Pipeline value
- Reservations
- Contracts
- Units sold
- Revenue
- Outstanding payments
- Upcoming meetings
- Open tasks
- Conversion rate
- Sales by owner
- Lead sources
- Recent activity

Dashboard widgets are configurable and link to source records.

6. CRM Module Header

- Add Client
- Add Client Company
- Add Contact
- Add Lead
- Add Opportunity

- Global Search
- Filters
- Saved Views
- Import
- Export
- List/Grid switch

7. Client Types

- Buyer
- Tenant
- Investor
- Company
- Government
- Partner
- Prospect
- Former Client
- VIP
- Custom

Client types are configurable and not hard-coded.

8. Client Statuses

- Draft
- Lead
- Prospect
- Qualified
- Active
- On Hold
- Completed
- Inactive
- Lost
- Archived

9. Client Cards and Rows

- Photo or logo
- Name
- Client ID
- Client type
- Status
- Pipeline stage
- Assigned salesperson
- Company
- Country
- Total purchases

- Outstanding balance
- Last activity
- Personal Star

Clicking a client opens the Client Page.

10. CRM Filters

- Client type
- Status
- Pipeline stage
- Assigned user
- Team
- Company
- Country
- Lead source
- Project of interest
- Unit type
- Opportunity value
- Outstanding balance
- Last activity
- Recently created
- Starred by me

11. Client Page Header

- Photo or logo
- Full name or company name
- Client code
- Client type
- Status
- Pipeline stage
- Assigned salesperson
- Assigned team
- Company
- Primary phone
- Primary email
- Preferred contact method
- Last activity

12. Client Page Actions

- Edit Client
- Create Task
- Schedule Meeting
- Upload Document
- Add Opportunity

- Add Reservation
- Create Contract
- Record Payment
- Add Note
- Add Support Case
- Archive

13. Individual Client Profile

- First name
- Middle name
- Last name
- Display name
- Date of birth
- Nationality
- Tax ID
- Identification/passport details
- Address
- Country
- Languages
- Phone numbers
- Email addresses
- Preferred contact method
- Preferred language
- Relationship status
- Internal notes

Sensitive identity fields must be permission-ready.

14. Client Company Profile

- Legal name
- Commercial name
- Registration number
- Tax/VAT number
- Industry
- Country
- Address
- Website
- Company size
- Client status
- Assigned account owner
- Primary contact
- Related contacts
- Related companies
- Contracts

- Opportunities
- Projects

Reuse the canonical Company entity when the client is an organization.

15. Contacts

- Contact name
- Position
- Department
- Company
- Phone
- Email
- Preferred method
- Decision influence
- Relationship owner
- Status
- Notes

One company may have multiple contacts. An individual client may also be a contact.

16. Contact Roles

- Decision maker
- Legal representative
- Finance contact
- Technical contact
- Sales contact
- Executive sponsor
- Assistant
- Family member
- Advisor
- Custom

Contact roles are configurable.

17. Leads

- Lead title
- Person or company
- Source
- Campaign
- Owner
- Team
- Interest
- Estimated value
- Status
- Priority
- Next action

- Expected qualification date
- Lost reason

18. Lead Sources

- Website
- Google
- Facebook
- Instagram
- Referral
- Walk-in
- Phone
- Event
- Exhibition
- Partner
- Existing client
- Custom

Lead sources are configurable and can be used in ROI reporting.

19. Lead Qualification

- Budget
- Need
- Authority
- Timeline
- Location preference
- Project interest
- Unit type
- Financing status
- Decision readiness
- Qualification score
- Qualification notes

Qualification logic should be transparent and configurable.

20. Opportunities

- Opportunity title
- Client
- Contact
- Project
- Units or services of interest
- Estimated value
- Probability
- Expected close date
- Owner
- Team

- Pipeline stage
- Competitors
- Risks
- Next action
- Status

21. Sales Pipeline

- Configurable stages
- Drag-and-drop board
- List and forecast views
- Stage probability
- Stage entry date
- Time in stage
- Required fields per stage
- Automation hooks later
- Lost and won reasons
- Pipeline filters

Pipeline stages are managed through Platform Configuration.

22. Default Pipeline Example

- Lead
- Contacted
- Qualified
- Meeting
- Site Visit
- Negotiation
- Reservation
- Contract
- Deposit Paid
- Completed
- After Sales

23. Reservation Management

- Client
- Project
- Building
- Unit
- Reservation date
- Expiration
- Deposit amount
- Deposit status
- Salesperson
- Reservation status

- Linked documents
- Linked approval
- Automatic release rules later

Unit reservation logic must remain synchronized with the Units Module.

24. Purchases and Sales

- Client
- Project
- Building
- Unit
- Asking price
- Discount
- Taxes
- Final sale price
- Sale date
- Salesperson
- Linked assets
- Contract
- Payment plan
- Sale status

Sale values must come from linked Unit, Contract and Finance records.

25. Unit Relationships

- Interested units
- Viewed units
- Reserved units
- Purchased units
- Rented units
- Released reservations
- Linked parking
- Linked storage
- Linked finish package

26. Contracts

- Reservation agreement
- Sale contract
- Rental contract
- Service agreement
- NDA
- Payment plan agreement
- Addendum
- Cancellation agreement
- Contract status

- Contract value
- Effective date
- Expiry
- Owner
- Linked documents

Contracts remain owned by the Contracts Module.

27. Payments and Financial Summary

- Contract value
- Amount paid
- Remaining balance
- Payment percentage
- Installment schedule
- Next payment
- Overdue payments
- Deposits
- Refunds
- Commissions
- Outstanding invoices
- Payment history

Financial values come from Finance. CRM displays filtered summaries and deep links.

28. Documents

- Identity documents
- Contracts
- Reservation documents
- Invoices
- Receipts
- Mortgage documents
- Bank documents
- Certificates
- Correspondence
- Support evidence

Uses the central Documents Module and Document Passport.

29. Tasks

- Client-related tasks
- List, Board, Calendar and Timeline shortcuts
- Participants
- Owner
- Priority
- Due date
- Status

- Linked opportunity
- Linked reservation
- Linked contract

Task visibility remains participant-based.

30. Meetings

- Upcoming meetings
- Previous meetings
- Site visits
- Sales meetings
- Contract meetings
- Support meetings
- Minutes
- Decisions
- Action items
- Linked documents and tasks

31. Communications

- Call log
- Email log
- WhatsApp log
- SMS log
- Internal message reference
- Communication direction
- Date/time
- User
- Summary
- Outcome
- Next action

Initial phase stores manual communication records; live integrations are future scope.

32. Notes

- Internal notes
- Rich text
- Mentions
- Attachments
- Tags
- Pinned notes
- Visibility readiness
- Author and timestamp

33. Approvals

- Discount approval

- Refund approval
- Payment plan approval
- Contract approval
- Reservation extension
- Special condition approval
- Status
- Approver
- Deadline
- Decision history

34. Client Timeline

- Lead created
- Contacted
- Meeting
- Site visit
- Opportunity created
- Reservation
- Contract
- Payment
- Unit sold
- Document uploaded
- Approval completed
- Support case
- Client status change

Every timeline event stores source module, actor, timestamp and linked record.

35. Support and After-Sales

- Support case
- Warranty issue
- Complaint
- Defect
- Maintenance request
- Priority
- Owner
- Status
- SLA target
- Linked unit/project
- Documents
- Tasks
- Resolution

36. Referrals

- Referrer

- Referred client
- Referral date
- Campaign
- Commission
- Reward
- Status
- Linked sale

37. Marketing Attribution

- Lead source
- Campaign
- Channel
- Ad/campaign identifier
- Cost allocation later
- Opportunities generated
- Revenue generated
- Conversion rate
- ROI reporting

No external advertising integration in the first phase.

38. Relationship Network

- Family members
- Client company
- Contact persons
- Salesperson
- Team
- Lawyer
- Bank
- Mortgage
- Projects
- Buildings
- Units
- Contracts
- Payments
- Meetings
- Tasks
- Documents
- Referrals
- Support cases

Render as an optional interactive graph while preserving normal table/list views.

39. CRM Reports

- Sales pipeline

- Conversion
- Revenue
- Lost leads
- Salesperson performance
- Team performance
- Lead sources
- Marketing attribution
- Client lifetime value
- Outstanding payments
- Reservations
- Contracts
- Support cases
- Client activity

40. CRM Settings

- Pipeline stages
- Lead sources
- Client types
- Client statuses
- Contact roles
- Lost reasons
- Opportunity probabilities
- Payment terms
- Numbering
- Communication templates
- Notification defaults
- Import mappings
- Archive rules

41. Platform Configuration Compatibility

- CRM module enable/disable
- Pages enable/disable
- Sections enable/disable
- Widgets enable/disable
- Features enable/disable
- Actions enable/disable
- Pipeline stages
- Client types
- Lead sources
- Reports
- Support module
- Marketing and referral sections

Disabled items must disappear from navigation, search, APIs and layouts without blank spaces or dead links.

42. Responsive UX

- Desktop: dashboard grid, pipeline board, split-view client list and detail.
- Tablet: two-column layout, collapsible side panels, horizontal pipeline.
- Mobile: compact client header, card lists, bottom action sheet, horizontal stage scroller.
- All core CRM actions remain available on mobile.

43. Premium Motion and Micro-interactions

- Client card hover: 120-180 ms lift, shadow and border tint.
- Pipeline drag: card elevation, placeholder and smooth stage transition.
- Stage progress: animated connector and chip transition.
- Timeline load: subtle staggered reveal.
- Payment progress: smooth bar interpolation.
- Relationship graph: restrained node expansion.
- Preview drawer: 220-280 ms slide and fade.
- Loading: skeletons matching final geometry.
- Respect reduced-motion settings.

44. Empty, Loading and Error States

- No clients
- No leads
- No opportunities
- No reservations
- No payments
- No communications
- No support cases
- Module disabled
- Source module unavailable
- Partial dashboard failure
- Archived client

45. Data Model

Entity	Purpose	Key fields
Client	Canonical client	id, tenant_id, type, status, display_name, owner_user_id, source_id, created_at
Person	Individual identity	client_id/contact_id, first_name, last_name, birth_date, nationality, tax_id
ClientCompany	Client-company relation	client_id, company_id, account_owner_id, status
Contact	Contact person	id, company_id, linked_client_id, name, title, department, email, phone, status
ContactRole	Configurable role	id, tenant_id, name, status
Lead	Pre-qualified interest	id, client_id/contact_id, source_id, owner_id, status, estimated_value, next_action_at

Entity	Purpose	Key fields
LeadSource	Configurable source	id, tenant_id, name, channel, status
Opportunity	Sales opportunity	id, client_id, project_id, owner_id, stage_id, value, probability, expected_close_at, status
Pipeline	Pipeline definition	id, tenant_id, name, status
PipelineStage	Configurable stage	pipeline_id, name, order_no, probability, required_fields_json, status
Reservation	Unit/service hold	id, client_id, unit_id, deposit_amount, reserved_at, expires_at, status
ClientPurchase	Purchase relation	id, client_id, unit_id/service_id, contract_id, final_value, sold_at, status
Communication	Manual communication log	id, client_id, type, direction, user_id, occurred_at, summary, outcome
ClientNote	Internal note	id, client_id, author_id, body, pinned, created_at
SupportCase	After-sales case	id, client_id, unit_id, type, priority, owner_id, status, opened_at, closed_at
Referral	Referral relation	id, referrer_client_id, referred_client_id, commission, status
CRMActivity	Timeline event	client_id, actor_id, source_module, entity_type, entity_id, event_type, occurred_at

46. Data Ownership Rules

- CRM owns clients, contacts, leads, opportunities, pipelines, notes, communications and support cases.
- Companies owns canonical organization records.
- Units owns unit availability and pricing.
- Contracts owns contract records.
- Finance owns invoices, payments and balances.
- Documents owns files and revisions.
- Tasks owns task visibility and execution.
- CRM summaries must be reproducible from source records.
- Deleting or archiving a CRM record must not delete source-module history.

47. API Requirements

- GET /crm/overview
- GET /clients
- POST /clients
- GET /clients/{id}
- PATCH /clients/{id}
- POST /clients/{id}/archive
- POST /clients/{id}/restore
- GET /clients/{id}/contacts
- GET /clients/{id}/opportunities
- GET /clients/{id}/reservations
- GET /clients/{id}/purchases
- GET /clients/{id}/contracts
- GET /clients/{id}/payments
- GET /clients/{id}/documents

- GET /clients/{id}/tasks
- GET /clients/{id}/meetings
- GET /clients/{id}/communications
- POST /clients/{id}/communications
- GET /clients/{id}/notes
- POST /clients/{id}/notes
- GET /clients/{id}/support-cases
- POST /clients/{id}/support-cases
- GET /clients/{id}/timeline
- GET /contacts
- POST /contacts
- GET /leads
- POST /leads
- GET /opportunities
- POST /opportunities
- PATCH /opportunities/{id}/stage
- GET /pipelines
- POST /pipelines
- GET /reservations
- POST /reservations
- GET /crm/reports

Aggregated endpoints return summaries and deep links rather than duplicated complete source records.

48. Frontend Architecture

- CRMModulePage
- CRMDashboard
- CRMFilterBar
- ClientList
- ClientCard
- ClientPage
- ClientHeader
- ClientOverview
- ClientContactsPanel
- ClientOpportunitiesPanel
- SalesPipelineBoard
- LeadPage
- OpportunityPage
- ReservationPanel
- PurchasesPanel
- PaymentsPanel
- ClientDocumentsPanel
- ClientTasksPanel
- ClientMeetingsPanel

- CommunicationTimeline
- ClientNotesPanel
- ClientApprovalsPanel
- SupportCasesPanel
- ReferralPanel
- RelationshipNetwork
- CRMReportsPage
- CRMSettingsPage

49. Backend Services and Background Jobs

- Client CRUD service
- Contact service
- Lead service
- Opportunity and pipeline service
- Reservation integration service
- Contract/finance summary resolver
- Communication log service
- Support case service
- Duplicate detection service
- Activity feed builder
- Search indexing service
- Report aggregation service
- Configuration resolver
- Cache invalidation service

Background jobs must be idempotent, tenant-aware and safe to retry.

50. Duplicate Detection and Merge

- Detect duplicate email, phone, tax ID, passport/ID number and company registration number.
- Warn before creating a likely duplicate.
- Allow controlled merge.
- Preserve source record IDs in merge history.
- Repoint linked records safely.
- Do not delete financial, contract or activity history.
- Record merge actor and timestamp.

51. Validation Rules

- Client display name required.
- At least one contact method may be required by configuration.
- Email and phone formats validated.
- Identification expiry must follow issue date.
- Opportunity value cannot be negative.
- Probability must be between 0 and 100.
- Expected close date validation.

- Reservation expiration must follow reservation date.
- Duplicate active reservation for the same unit prevented.
- Pipeline stage transitions validate required fields.
- Archive requires impact confirmation.
- Archived clients cannot receive new sales records unless restored.

52. Security and Permission Readiness

- Tenant isolation
- Company scope validation
- Sensitive identity fields separated
- Financial summaries access-ready
- Internal notes visibility-ready
- Read/write audit for sensitive fields
- No linked module access expansion through CRM
- Section-level, field-level and action-level permission readiness
- Platform Configuration and permissions evaluated separately

53. Performance and Scalability

- Load dashboard and client header first.
- Lazy-load client sections.
- Cursor pagination for clients, activity and communications.
- Virtualize pipeline boards and large client lists.
- Avoid N+1 module aggregation queries.
- Precompute dashboard summaries.
- Incrementally update pipeline counters.
- Support millions of clients, contacts and activities.
- Target initial interactive load below 2 seconds under normal enterprise conditions.

54. Testing Strategy

Unit tests

- Client validation
- Duplicate detection
- Pipeline transitions
- Opportunity probability
- Reservation validation
- Merge logic
- Archive/restore
- Platform Configuration resolution

Integration tests

- Company/contact linkage
- Units reservation
- Contracts
- Finance summaries

- Documents
- Tasks privacy
- Meetings
- Activity feed

End-to-end tests

- Create individual client
- Create company client and contacts
- Create lead
- Qualify lead
- Create opportunity
- Move through pipeline
- Reserve unit
- Create contract
- Record payment summary
- Create support case
- Merge duplicates
- Disable CRM section through Platform Configuration
- Mobile behavior

Visual regression

- CRM dashboard
- Client Page
- Pipeline board
- Tablet
- Mobile
- Large client list
- Disabled-section reflow
- Reduced-motion mode

55. Acceptance Criteria

- AC-01 - The CRM supports individual clients and client companies.
- AC-02 - A company client can have multiple contacts.
- AC-03 - Leads can be qualified and converted to clients/opportunities.
- AC-04 - Pipeline stages are configurable.
- AC-05 - Opportunities support value, probability and expected close date.
- AC-06 - Reservations synchronize with Units.
- AC-07 - Contracts and payments link to source modules.
- AC-08 - Documents use the central Documents Module.
- AC-09 - Tasks preserve participant-based visibility.
- AC-10 - Communications can be logged manually.
- AC-11 - Client timeline aggregates all major events.
- AC-12 - Support and after-sales cases are supported.
- AC-13 - Duplicate detection and controlled merge are available.
- AC-14 - Relationship Network is available as an optional view.

- AC-15 - CRM reports and attribution are supported.
- AC-16 - Desktop, tablet and mobile retain core actions.
- AC-17 - Premium animations are implemented.
- AC-18 - Reduced-motion preferences are respected.
- AC-19 - Every CRM element supports Platform Configuration.
- AC-20 - Disabled features leave no dead links or blank spaces.
- AC-21 - Large CRM datasets remain performant.

56. Page and Flow Registry

- CRM Module Page
- CRM Dashboard
- Clients
- Client Page
- Create Client
- Edit Client
- Archive Client
- Restore Client
- Client Companies
- Contact Page
- Create Contact
- Leads
- Lead Page
- Create Lead
- Lead Qualification Flow
- Convert Lead Flow
- Opportunities
- Opportunity Page
- Create Opportunity
- Sales Pipeline
- Pipeline Stage Setup
- Reservation Flow
- Client Purchases
- Client Units
- Client Contracts
- Client Payments
- Client Documents
- Client Tasks
- Client Meetings
- Client Communications
- Client Notes
- Client Approvals
- Client Timeline
- Support Cases

- Referrals
- Marketing Attribution
- Relationship Network
- CRM Reports
- CRM Settings
- Duplicate Merge Flow

57. Implementation Phases

- Phase 1: Client, contact, company-client, lead and pipeline foundations.
- Phase 2: Opportunities, CRM dashboard, Client Page and responsive UI.
- Phase 3: Reservations, purchases, units, contracts and finance summaries.
- Phase 4: Documents, tasks, meetings, communications, notes and approvals.
- Phase 5: Support, referrals, marketing attribution, reports and relationship graph.
- Phase 6: Performance, audit, accessibility, configuration and test hardening.

58. Future Roadmap

- Client Portal
- Email integration
- WhatsApp integration
- SMS integration
- Telephony integration
- Digital signatures
- Marketing automation
- AI lead scoring
- AI communication summaries
- AI next-best-action suggestions

Future integrations are not part of the initial implementation.

59. Final Architectural Rule

The CRM Module must remain a universal, configurable and source-linked relationship engine. It must support any business category, preserve data ownership across Units, Contracts, Finance, Documents and Tasks, and remain fully compatible with Platform Configuration and future permissions.